Lecture 8. Numerical Integration

Newton–Cotes rules, Gaussian quadrature, orthogonal polynomials, Laplace approximation

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We want $\int_a^b f(x)\,dx$. Integrals are everywhere in statistics but rarely have closed forms:

- expectations, $E\sin X = \int \sin(x) f_X(x)\,dx$;
- Bayesian normalizing constants $\int f(x \mid \theta)\pi(\theta)\,d\theta$;
- likelihoods of mixed models, e.g. a Poisson GLM with random intercept, $\prod_i \int \prod_j f(y_{ij} \mid \gamma_i)\phi(\gamma_i)\,d\gamma_i$.

Two approaches: quadrature (now) and Monte Carlo (Lecture 10).

8.1 Newton–Cotes rules

Split $[a, b]$ into n intervals of length $h = (b-a)/n$, $x_i = a + ih$, approximate f on each $[x_i, x_{i+1}]$ by an interpolating polynomial of degree m, integrate the polynomial exactly, and add up.

- $m = 0$, **Riemann**: $\int_{x_i}^{x_{i+1}} f \approx h f(x_i)$.
- $m = 1$, **trapezoidal**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{2}\{f(x_i) + f(x_{i+1})\}$, so $T(n) = h\{\frac{1}{2}f(a) + f(x_1) + \cdots + f(x_{n-1}) + \frac{1}{2}f(b)\}$.
- $m = 2$, **Simpson**: $\int_{x_i}^{x_{i+1}} f \approx \frac{h}{6}\{f(x_i) + 4f(\frac{x_i + x_{i+1}}{2}) + f(x_{i+1})\}$.

Proposition 8.1 (Accuracy)

For smooth f, the total errors are $O(n^{-1})$ (Riemann), $O(n^{-2})$ (trapezoidal) and $O(n^{-4})$ (Simpson). More precisely, per subinterval the local errors are

$$\frac{h^2}{2}f'(x_i) + O(h^3), \qquad -\frac{h^3}{12}f''(x_i) + O(h^4), \qquad -\frac{h^5}{2880}f^{(4)}(x_i) + O(h^6).$$

Proof. Integrate the Taylor expansion $f(x) = f(x_i) + (x - x_i)f'(x_i) + \frac{1}{2}(x - x_i)^2 f''(x_i) + \ldots$ over $[x_i, x_i + h]$:

$$\int_{x_i}^{x_{i+1}} f = h f(x_i) + \frac{h^2}{2}f'(x_i) + \frac{h^3}{6}f''(x_i) + \frac{h^4}{24}f^{(3)}(x_i) + \frac{h^5}{120}f^{(4)}(x_i) + O(h^6).$$

Riemann: the error is $\frac{h^2}{2}f'(x_i) + O(h^3)$. *Trapezoid*: expanding $f(x_{i+1})$ gives $\frac{h}{2}\{f(x_i) + f(x_{i+1})\} = h f(x_i) + \frac{h^2}{2}f' + \frac{h^3}{4}f'' + O(h^4)$, so the error is $(\frac{1}{6} - \frac{1}{4})h^3 f'' + O(h^4) = -\frac{h^3}{12}f''(x_i) + O(h^4)$. *Simpson*: expanding f at the midpoint and endpoint, the rule matches the terms through h^4 and differs at h^5 by $(\frac{1}{120} - \frac{5}{576})h^5 f^{(4)} = -\frac{h^5}{2880}f^{(4)}$. Summing n local errors multiplies by $n = (b-a)/h$, giving global orders h, h^2, h^4. ∎

Simpson uses about twice as many function evaluations per interval, so a fair comparison fixes the total number of evaluations; Simpson still wins by orders of magnitude for smooth f.

General m: interpolation. With $m+1$ distinct nodes $x_0, \ldots, x_m$, the **Lagrange polynomials** $L_j(x) = \prod_{k \neq j} \frac{x - x_k}{x_j - x_k}$ satisfy $L_j(x_k) = \delta_{jk}$, so $p(x) = \sum_j y_j L_j(x)$ interpolates (x_j, y_j).

Theorem 8.2 (Polynomial interpolation)

For $m + 1$ distinct points there is a unique polynomial of degree at most m through $(x_0, y_0), \ldots, (x_m, y_m)$.

Proof. Existence is the Lagrange form. If q is another, $p - q$ has degree $\leq m$ and $m + 1$ roots, hence is identically 0. ∎

Corollary 8.3

If f is a polynomial of degree at most m, then $\int_a^b f = \sum_j w_j f(x_j)$ with $w_j = \int_a^b L_j(x)\,dx$.

Proof. By uniqueness, $f = \sum_j f(x_j)L_j$; integrate. ■

8.2 Gaussian quadrature

Newton–Cotes fixes equally spaced nodes and chooses weights to integrate polynomials of degree m exactly. Choosing the nodes as well gains much more.

Polynomials of degree $\le m$ on (a, b) form a vector space $\mathcal{P}^m$ with inner product

$$\langle p, q\rangle = \int_a^b p(x)q(x)w(x)\,dx$$

for a positive weight w (here $w \equiv 1$). Gram–Schmidt applied to $1, x, x^2, \ldots$ produces orthonormal polynomials $q_0, q_1, \ldots$, with q_k of exact degree k and orthogonal to all of $\mathcal{P}^{k-1}$; for $w \equiv 1$ on $(-1, 1)$ these are the **Legendre polynomials**.

Lemma 8.4

The roots of q_k are real, simple and lie in (a, b).

Proof. Let $r_1 < \cdots < r_l$ be the points in (a, b) where q_k changes sign. If $l < k$, then $q_k(x)\prod_{i\le l}(x-r_i)$ does not change sign on (a, b) and is not identically zero, so its integral is nonzero. But $\prod_{i\le l}(x-r_i)$ has degree $l < k$ and is orthogonal to q_k: a contradiction. So q_k has k sign changes, hence k distinct real roots in (a, b). ■

Theorem 8.5 (Gaussian quadrature)

Let $x_0, \ldots, x_m$ be the roots of q_{m+1} and $w_j = \int_a^b L_j(x)w(x)\,dx$. Then

$$\int_a^b f(x)w(x)\,dx = \sum_{j=0}^{m} w_j f(x_j)$$

for every polynomial f of degree at most $2m + 1$, and all $w_j > 0$. Degree $2m + 1$ cannot be improved.

Proof. Divide: $f = gq_{m+1} + r$ with $g, r \in \mathcal{P}^m$. Then $\int gq_{m+1}w = \langle g, q_{m+1}\rangle = 0$, so $\int fw = \int rw$. At the nodes $q_{m+1}(x_j) = 0$, so $f(x_j) = r(x_j)$, and by Corollary 8.3 (with weight w) $\int rw = \sum w_j r(x_j) = \sum w_j f(x_j)$. Positivity: L_k^2 has degree $2m$, so $0 < \int L_k^2 w = \sum_j w_j L_k(x_j)^2 = w_k$. Optimality: $f = \prod_j (x - x_j)^2$ has degree $2m + 2$ and $\int fw > 0$, while the rule gives 0. ■

So $m + 1$ well-chosen nodes integrate polynomials of degree $2m + 1$ exactly, against degree m for equally spaced nodes.

Weight functions. Changing the weight gives different families of orthogonal polynomials, suited to different integrals $\int g(x)w(x)\,dx$:

polynomials	weight $w(x)$	domain	related distribution
Legendre	1	$(-1, 1)$	uniform
Laguerre	$x^\alpha e^{-x}$	$(0, \infty)$	gamma
Hermite	e^{-x^2}	$(-\infty, \infty)$	normal
Jacobi	$(1 - x)^\alpha(1 + x)^\beta$	$(-1, 1)$	beta (after rescaling)

Each family satisfies a three-term recursion $q_m(x) = (\alpha_m + \beta_m x)q_{m-1}(x) - \gamma_m q_{m-2}(x)$, and its roots *interlace* (the roots of q_{m+1} separate those of q_m). The nodes are computed by Newton's method using these properties (or as eigenvalues of a tridiagonal matrix).

Gauss–Hermite quadrature. $\int f(x)\,dx = \int \{f(x)e^{x^2}\}e^{-x^2}dx \approx \sum_j w_j f(x_j)e^{x_j^2}$. It works well only if $f(x)e^{x^2}$ is close to a polynomial. For $f(x) = e^{-(x-\mu)^2/(2\sigma^2)}$ with μ far from 0 or σ^2 far from $\frac{1}{2}$, the nodes sit where f has no mass and the rule fails. The remedy is to *centre and scale*: substituting $z = (x-\mu)/(\sqrt{2}\sigma)$ gives $\sqrt{2}\sigma \int e^{-z^2}dz$, which a single node integrates exactly.

Example 8.1 (Binomial–logit posterior)

$x \sim B(n,p)$ with $p = e^\theta/(1+e^\theta)$ and prior $\theta \sim N(0,\sigma^2)$. The posterior normalizing constant is $\int \bar{g}(\theta)\,d\theta$ with $\bar{g}(\theta) = e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$. Applying Gauss–Hermite after only rescaling by σ performs badly when $x/n = 0.9$, because the integrand peaks near $\theta = \log 9 = 2.2$, far from the nodes. Centring at the maximizer θ^* of $\bar{g}$ and writing $\int \bar{g} = \int \{\bar{g}(z+\theta^*)e^{z^2}\}e^{-z^2}dz$ gives an accurate rule with few nodes.

8.3 Laplace approximation

For integrals $\int f(x)e^{-ag(x)}dx$ with large a, where g has a unique minimum at 0, the integrand concentrates near 0, and $g(x) \approx g(0) + \frac{1}{2}g''(0)x^2$ gives

$$\int f(x)e^{-ag(x)}dx \approx f(0)e^{-ag(0)}\sqrt{\frac{2\pi}{ag''(0)}}:$$

integration is replaced by optimization.

Theorem 8.6 (Laplace)

Suppose (1) for every $\delta > 0$ there is $\rho > 0$ with $g(x) - g(0) \geq \rho$ for $|x| \geq \delta$; (2) g is twice continuously differentiable near 0 with $g'(0) = 0$, $g''(0) > 0$; (3) f is continuous near 0 with $f(0) > 0$; (4) $\int |f|e^{-ag} < \infty$ for $a \geq a_0$. Then

$$\lim_{a\to\infty} \frac{\int f(x)e^{-ag(x)}dx}{f(0)e^{-ag(0)}\sqrt{2\pi/(ag''(0))}} = 1.$$

Proof. Take $g(0) = 0$. Given $\epsilon \in (0, \frac{1}{2}g''(0))$, choose δ such that for $|x| < \delta$,

$$(1-\epsilon)f(0) \leq f(x) \leq (1+\epsilon)f(0), \qquad \tfrac{1}{2}(g''(0)-2\epsilon)x^2 \leq g(x) \leq \tfrac{1}{2}(g''(0)+2\epsilon)x^2.$$

Tails. For $|x| \geq \delta$, $g \geq \rho$, so for $a \geq a_0$,

$$\int_{|x|\geq\delta} |f|e^{-ag} = \int_{|x|\geq\delta} |f|e^{-a_0 g}e^{-(a-a_0)g} \leq e^{-(a-a_0)\rho}\int |f|e^{-a_0 g} = O(e^{-\rho a}).$$

Centre, upper bound. $\int_{|x|<\delta} fe^{-ag} \leq (1+\epsilon)f(0)\int e^{-\frac{a}{2}(g''(0)-2\epsilon)x^2}dx = (1+\epsilon)f(0)\sqrt{\frac{2\pi}{a(g''(0)-2\epsilon)}}$. *Centre, lower bound.* $\int_{|x|<\delta} fe^{-ag} \geq (1-\epsilon)f(0)\int_{|x|<\delta} e^{-\frac{a}{2}(g''(0)+2\epsilon)x^2}dx$, and the Gaussian integral over $|x| \geq \delta$ is $O(e^{-\lambda a})$ with $\lambda = \frac{1}{2}(g''(0)+2\epsilon)\delta^2$. Multiplying by $\sqrt{a}$ and letting $a \to \infty$,

$$(1-\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)+2\epsilon}} \leq \liminf \sqrt{a}\int fe^{-ag} \leq \limsup \sqrt{a}\int fe^{-ag} \leq (1+\epsilon)f(0)\sqrt{\frac{2\pi}{g''(0)-2\epsilon}}.$$

Let $\epsilon \to 0$. ∎

Example 8.2 (Stirling's formula)

With $x = tz$, $\Gamma(t+1) = \int_0^\infty x^t e^{-x} dx = t^{t+1}\int_0^\infty e^{-t(z-\log z)}dz$. Here $g(z) = z - \log z$ is minimized at $z = 1$ with $g(1) = 1$, $g''(1) = 1$, so $\Gamma(t+1) \sim \sqrt{2\pi}\, t^{t+1/2}e^{-t}$.

Posterior expectations. With log-likelihood $\ell_n(\theta)$, log-prior $\pi(\theta)$ and $\tilde{h} = \log h > 0$,

$$E[h(\Theta) \mid x] = \frac{\int e^{\tilde{h}+\ell_n+\pi} d\theta}{\int e^{\ell_n+\pi} d\theta}.$$

Apply Laplace to numerator and denominator separately, at the maximizers $\tilde{\theta}$ of $\tilde{h}+\ell_n+\pi$ and $\hat{\theta}$ of $\ell_n+\pi$ (the posterior mode):

$$E[h(\Theta) \mid x] \approx \exp\{(\tilde{h}+\ell_n+\pi)(\tilde{\theta}) - (\ell_n+\pi)(\hat{\theta})\}\sqrt{\frac{(\ell_n+\pi)''(\hat{\theta})}{(\tilde{h}+\ell_n+\pi)''(\tilde{\theta})}}.$$

For large n the log-posterior is sharply peaked, and the approximation is accurate (its relative error is $O(n^{-2})$, since the leading errors of numerator and denominator cancel).

Example 8.3 (Beta moments)

For $\Theta \sim \text{Beta}(\alpha, \beta)$ (a binomial posterior), $E\Theta^k = \frac{\alpha(\alpha+1)\cdots(\alpha+k-1)}{(\alpha+\beta)\cdots(\alpha+\beta+k-1)}$ exactly. With $g(\theta) = -(\alpha-1)\log\theta - (\beta-1)\log(1-\theta)$, minimized at $\hat{\theta} = \frac{\alpha-1}{\alpha+\beta-2}$,

$$\int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}d\theta \approx \Big(\frac{\alpha-1}{\alpha+\beta-2}\Big)^{\alpha-1}\Big(\frac{\beta-1}{\alpha+\beta-2}\Big)^{\beta-1}\sqrt{\frac{2\pi(\alpha-1)(\beta-1)}{(\alpha+\beta-2)^3}},$$

and the same with α replaced by $\alpha+k$ for the numerator. If $\alpha, \beta \to \infty$ with $\alpha/(\alpha+\beta) \to p$, both the exact moment and its approximation tend to p^k.

Unlike quadrature, whose number of nodes grows exponentially with dimension, the Laplace approximation extends directly to $\mathbb{R}^d$: $\int f e^{-ag} \approx f(\hat{x})e^{-ag(\hat{x})}(2\pi/a)^{d/2}\det\{\nabla^2 g(\hat{x})\}^{-1/2}$.